

Observed-data efficiency for a sharp smooth conditional-transport endpoint

CausalSmith

Locked parameters. The observed data are i.i.d. triangular-array draws $O_{i,n} = (X_{i,n}, A_{i,n}, Y_{i,n})$, where $A_{i,n} \in \{0, 1\}$. The target is

$$\theta_n = \mathbb{E}_{P_n} [W_2^2\{Q_{0,P_n}(\cdot | X_n), Q_{1,P_n}(\cdot | X_n)\}],$$

the sharp conditional-coupling lower endpoint on the mean squared individual treatment effect under consistency and conditional ignorability. The regular regime has fixed overlap, compact support, bounded Hölder conditional arm densities, strong Brenier regularity, and efficiency variance separated from zero.

Abstract

The sharp lower endpoint on mean squared individual treatment effects is an observed-law conditional optimal-transport functional once consistency and conditional ignorability identify each potential-outcome kernel with an observed treatment-arm kernel. Theorem 1 gives its observed-data canonical gradient and semiparametric efficiency bound. Conjecture 2, explicitly not yet proved, specifies a normalized, foldwise coupled density–transport–propensity construction and a conditional Hölder/DML route to uniform studentized Wald inference in the regular regime. The density route derives the transport learner rates from smoothness, while the propensity rate is separately required. Equality of the two conditional laws and the resulting vanishing first-order variance remain outside the regular result.

1 Introduction

Individual treatment effects are not observed jointly, so their mean squared magnitude is generally only partially identified. Conditional coupling of the two potential-outcome laws gives the sharp lower endpoint, and quadratic conditional optimal transport expresses that endpoint as a smooth functional when the conditional arm laws are regular.

Lin, Gao, Blanchet, and Glynn (?) study this sharp smooth conditional-transport endpoint with randomized marginal samples. Their Theorem 3.2 is a pointwise quadratic-COT central limit theorem, Proposition 3.3 gives the quadratic stability expansion, Proposition 3.2 controls integrated conditional W_2^2 estimation, and Section C.4 proves the limit theorem. Those results do not differentiate the endpoint along observational treatment-assignment, outcome-kernel, and covariate-margin tangent directions. The question is whether that observed-law derivative supports efficient, uniformly studentized inference for the sharp primal endpoint.

The immediate consumer is a sharp lower confidence interval for individual-effect heterogeneity in the 401(k) SIPP setting studied by ?. A practitioner could use a two-sided efficient interval for the sharp lower endpoint when the regular regime is credible, rather than a conservative one-sided dual-bound procedure.

Results at a glance.

- **Theorem 1:** the observed-law endpoint has a canonical gradient and efficiency bound. Gap: ? is randomized-sample rather than observational. Consumer: efficiency benchmarking for the 401(k) SIPP endpoint.
- **Conjecture 2 (not yet proved):** a normalized coupled one-step estimator is uniformly efficient and studentized-normal under the stated conditional DML rates. Gap: ? gives observational dual-bound AIPW inference, not sharp-primal two-sided efficient Wald inference. Consumer: regular-regime sharp Wald intervals.
- **Open-ended Question 1:** find an adaptive procedure when the first-order efficiency variance vanishes. Gap: existing transport degeneracy results do not include conditional transport and observational scores. Consumer: inference near equality of the arm kernels.

2 Related work

? provide the closest sharp-primal randomized-sample analysis; their cited results delimit the random-design, pointwise guarantee from the observational and uniform conjecture here. ? establish weak and strong duality in Section 2.1 and Theorem 2.1(1), fitted dual bounds in equations (11)–(13), and an observational AIPW extension in Section 3.4, equation (26), and Theorem 3.5. Their dual-bound procedure can retain weak-duality slack and does not supply the sharp-primal efficient score.

?, Section 3.1 and Theorem 6, provide the smooth strongly-convex Brenier stability inequality and a quadratic unconditional W_2^2 remainder. Applying that logic pointwise in covariates and jointly to two fitted marginals is a required, unproved step of Conjecture 2. ? and ? provide the cross-fitting and orthogonal-score ancestors, not the conditional-transport derivative. The in-repository proposal `stat_prte_ot_boundary_adaptation_v1` concerns deteriorating-overlap boundary remainders; fixed overlap is maintained here.

3 Formal setup

Let $Q_{a,P_n}(B \mid x) = P_n(Y_n \in B \mid A_n = a, X_n = x)$ and suppose it has density $q_{a,n}(x, y)$ on compact $\mathcal{Y}$. Define

$$V_n(x) = W_2^2\{Q_{0,P_n}(\cdot \mid x), Q_{1,P_n}(\cdot \mid x)\}, \quad \theta_n = \mathbb{E}_{P_n}[V_n(X_n)].$$

Consistency and ignorability identify $Q_{a,P_n}(\cdot \mid x)$ with $\mathcal{L}(Y_n(a) \mid X_n = x)$. Let $u_{a,n}$ be conditional quadratic dual potentials and center them as

$$\phi_{a,n}(x, y) = u_{a,n}(x, y) - \int u_{a,n}(x, z) Q_{a,P_n}(dz \mid x).$$

The candidate observed-data gradient is

$$\psi_n(O) = V_n(X) - \theta_n + \sum_{a=0}^1 \frac{\mathbf{1}\{A = a\} \phi_{a,n}(X, Y)}{\pi_{a,n}(X)}.$$

4 Assumptions

- A1.** $O_{1,n}, \dots, O_{n,n}$ are i.i.d. from P_n , consistency and conditional ignorability hold, and $\varepsilon \leq \pi_n(X_n) \leq 1 - \varepsilon$ almost surely.
- A2.** $\mathcal{X}$ and $\mathcal{Y}$ are compact; $c \leq q_{a,n}(x, y) \leq C$, $q_{a,n} \in \Sigma(s, L; \mathcal{X} \times \mathcal{Y})$, and the centered conditional dual potentials have the stated λ -strong Brenier regularity.
- A3.** The regular class is variance separated: $\sigma_n^2 = \mathbb{E}_{P_n}[\psi_n(O)^2] \geq \sigma_\star^2 > 0$.

5 Main results

Theorem 1 (Observed-data gradient and efficiency bound). *Under A1, $Q_{a,P_n}(\cdot | x) = \mathcal{L}(Y_n(a) | X_n = x)$ for almost every x . Over the regular observed-data class, ψ_n is the canonical gradient of θ_n in the nonparametric observed-data model, and σ_n^2 is its semiparametric efficiency bound.*

Conjecture 1 (Regular efficient inference, not yet proved). *Partition the sample into fixed K folds. On each training complement, fit a jointly measurable constrained conditional-density wavelet or sieve $\hat{q}_{a,-k,n}$ with $c \leq \hat{q}_{a,-k,n} \leq C$ and $\int \hat{q}_{a,-k,n}(x, y) dy = 1$, and clip the fitted propensity to $[\varepsilon, 1 - \varepsilon]$. Use a deterministic minimum-norm normalized measurable selection of the conditional quadratic coupling and dual potentials for this same fitted pair. Center each fitted potential under its own fitted kernel and define the held-out score*

$$\hat{S}_{-k,n}(O) = \hat{V}_{-k,n}(X) + \frac{A\hat{\phi}_{1,-k,n}(X, Y)}{\bar{\pi}_{-k,n}(X)} + \frac{(1 - A)\hat{\phi}_{0,-k,n}(X, Y)}{1 - \bar{\pi}_{-k,n}(X)}.$$

Put $D = d_X + d_Y$, use $2^{J_n D} \asymp n^{D/(2s+D)}$, and suppose $s > D/2$. The density route has $r_{q,n} = n^{-s/(2s+D)}$; require separately $r_{\pi,n} = O(n^{-\beta})$ with $\beta + s/(2s + D) > 1/2$. Then, uniformly over the regular class, the coupled construction has the exact conditional bias identity

$$\mathcal{B}_{k,n} = \mathbb{E}_{P_{X,n}} \left[\hat{V}_{-k,n} - V_n + \sum_a \int \hat{\phi}_{a,-k,n} d(Q_{a,P_n} - \hat{Q}_{a,-k,n}) + \sum_a \left\{ \frac{\pi_{a,n}}{\bar{\pi}_{a,-k,n}} - 1 \right\} \int \hat{\phi}_{a,-k,n} d(Q_{a,P_n} - \hat{Q}_{a,-k,n}) \right]$$

The proposed stability argument yields $\mathcal{B}_{k,n} = O_P(r_{q,n}^2 + r_{\pi,n} r_{q,n}) = o_P(n^{-1/2})$, derived L^2 rates $r_{\phi,n} = r_{V,n} = O(r_{q,n})$, asymptotic linearity with gradient ψ_n , consistent positive-root empirical studentization, uniform normality, and two-sided Wald coverage.

Open-ended Question 1 (Degeneracy-adaptive inference). *Determine a normalization, limit law, and data-driven confidence procedure with uniform coverage along observational sequences for which $\sigma_n \rightarrow 0$, including equality $Q_{0,P_n}(\cdot | X_n) = Q_{1,P_n}(\cdot | X_n)$ almost surely. The handle is a second conditional Brenier derivative along local arm-kernel perturbations combined with derivative-aware resampling of the observed-data score.*

6 Sanity-check corollaries

If $\pi_n(X_n) = 1/2$ almost surely, the candidate gradient reduces algebraically to the equal-probability observational score. This is a score check only; it is

not a fixed-balanced-randomization theorem.

7 Proof-sketch route

Theorem 1 follows by identifying the observed arm kernels with potential-outcome kernels, differentiating the conditional quadratic value in each arm-kernel direction, and projecting the derivative into the observed-data tangent space. For Conjecture 2, constrained normalization makes each fitted object a probability kernel; measurable selection fixes the conditional optimizer; fitted centering gives the displayed bias identity; smooth two-marginal stability gives its quadratic primal term and potential/value rates; clipping and Cauchy–Schwarz give the propensity–kernel product; cross-fitting then turns the expansion into uniform studentized inference. The last conditional stability and uniform empirical-process steps remain unproved.

8 Honest scope flags

Theorem 1 is retained. Conjecture 2 is not yet proved and is conditional on the separately attained propensity rate; it excludes deteriorating overlap, support-boundary remainders, equality, and vanishing variance. The open-ended question is the sole degeneracy frontier.

9 Iteration trail

- v1 specified the observed-law endpoint and causal identification.
- v2 coupled fitted arm kernels, propensity clipping, values, and centered duals.
- v3 retained the canonical-gradient theorem and isolated degeneracy as an open question.
- v4 relabeled regular inference as Conjecture 2 and corrected the Lin et al. record to the arXiv preprint.
- v5 corrected the Manole et al. citation to *Annals of Statistics* 52(3): 966–998.
- v6 normalized fitted kernels, specified measurable OT selection, added the conditional Hölder/DML rate route, verified source anchors, and synchronized this rendered proposal with the core.

References

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